



The Dartmouth workshop

A fascinating first person account of what took place at the workshop where modern AI was born. <https://digit.in/jan20-79>



Existential risk of AI

Find out exactly why AI is considered to be a threat looming over mankind, one that could destroy us all. <https://digit.in/jan20-80>

'deep learning' during this period which allowed computers to learn with experience. However, ambitious goals during this phase weren't met and the field experienced yet another 'Winter' from approximately 1988 to 1993 which coincided with the collapse of the market for general-purpose computers.

THE REVIVAL OF AI

Despite the lack of government funding and public hype, AI research and innovations continued to thrive. Subsequently, in the 1990s, American corporations' interest in AI applications was revitalised. In 1995, Richard Wallace, a computer scientist, developed the AI chatbot A.L.I.C.E. (Artificial Linguistic Internet Computer Entity), largely inspired by ELIZA by Weizenbaum. However, the difference was the addition of natural language sample datasets.

The most celebrated development in this century was a chess-playing computer program by IBM dubbed 'Deep Blue' beating reigning world

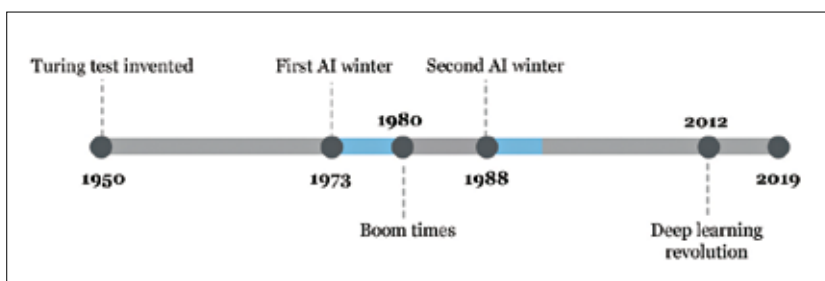


'Deep Blue', an AI computer program beat the world chess champion in 1997

chess champion and grand master, Gary Kasparov at the game in the year 1997. The match was highly publicised and this pushed AI into the limelight. AI was also becoming more capable of handling NLP and recognising and displaying human-like emotions.

Even though AI funding saw a slight setback in the early 2000s due to the dotcom bubble burst and issues due to Y2K, ML kept going strong owing to improvements in computer hardware. Governments and corporations were successfully utilising ML methods in narrow domains at this time.

The fundamental limit of computer storage which held AI back 30 years ago



AI Winters timeline

was no longer an issue. By this time, the memory and speed of computers managed to catch up, and in some cases, even surpass the needs of AI applications. Computational power was the primary obstacle and this issue was getting less pressing in the 2000s. Today, the issue has been vanquished for the most part.

We saw innovations such as Kismet, a robot that could recognise and relay emotions on its face (2000); ASIMO by Honda which was an AI humanoid robot (2000); iRobot's Roomba, an autonomous robot vacuum that can avoid obstacles (2002); NASA's robotic exploration rovers, Spirit and Opportunity, which navigated through the surface of Mars without human intervention (2004); and more. In 2007, Fei Fei Li (computer science professor) and his colleagues created ImageNet. This is a database of annotated images which could aid in object recognition software research. Google was secretly developing a driverless car at the end of this decade, which passed Nevada's self-driving test in 2014.

AI IN THE 2010S AND THE POSSIBLE FUTURE

The 2010s was an extremely significant decade for AI innovation. AI began to become embedded in our day-to-day lives more than ever and devices we carry around every day i.e. our smartphones, tablets and smartwatches began having AI ingrained in them. With the escalation of 'big data', the process of collecting large sums of data and processing it became too cumbersome for humans. AI has been of huge assistance in this regard and began being used fruitfully in industries such as banking, marketing, and technology.

Even though algorithms didn't improve substantially, big data and massive computing abilities allowed AI to learn more through brute force. Tech moguls such as Google, Amazon, Baidu and others began storing vast quantities of data for the very first time. They leveraged ML to gain massive commercial advantages by processing user data to understand consumer behaviour and align their marketing and recommendations according to these observations. NLP and other AI applications also became prevalent, while ML is embedded in almost all the online services we use. Voice Assistants such as Siri, Google Assistant, Amazon Alexa and more used NLP to infer, observe, answer and recommend to the human user. Anthropomorphised to create a better user experience, voice assistants began offering hyper-individualised experiences.

So, what's the future of AI going to look like? Language processing in AI can still go a long way and applications such as Google Assistant being able to make reservations for you should become more commonplace. Driverless cars is yet another avenue that is continually seeing developments and obstacles alike, however, the next decade could see the proliferation of these in a couple of cities in developed countries. The long-term goal seems to be creating machines that can surpass human cognitive abilities in almost every task. While this will still take much more than a decade, ethical concerns regarding AI could serve as barriers in this regard. Conversations and regulations regarding machine policy and ethics should steadily increase, however, one thing is for sure. AI is not going anywhere and with time, its applications will only mushroom. 